SYSADMIN MANUAL · ENTERPRISE EDITION

PrathamOne EduOS

SysAdmin Security & Architecture Manual

> Comprehensive operational, administrative, and technical manual for the complete coaching institute lifecycle management platform.

SOFTWARE PLATFORM
PrathamOne Coaching Institutes Enterprise Suite

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OFFICIAL WEBSITE
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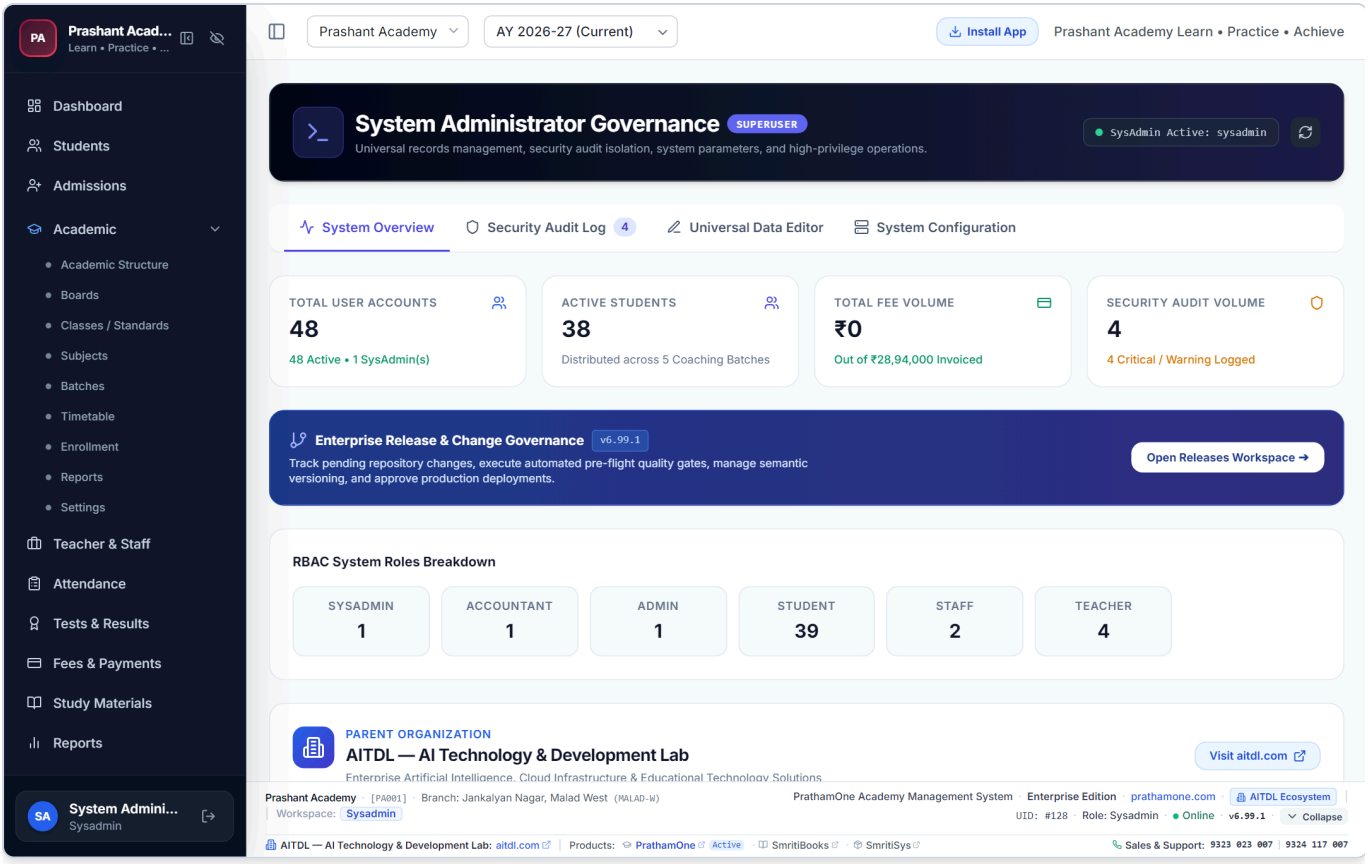
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1. SysAdmin Superuser Privileges & Architecture

The **SysAdmin** role represents the highest level of administrative authorization within the PrathamOne EduOS enterprise architecture. SysAdmin accounts possess unrestricted visibility across all application modules, as well as exclusive capabilities:

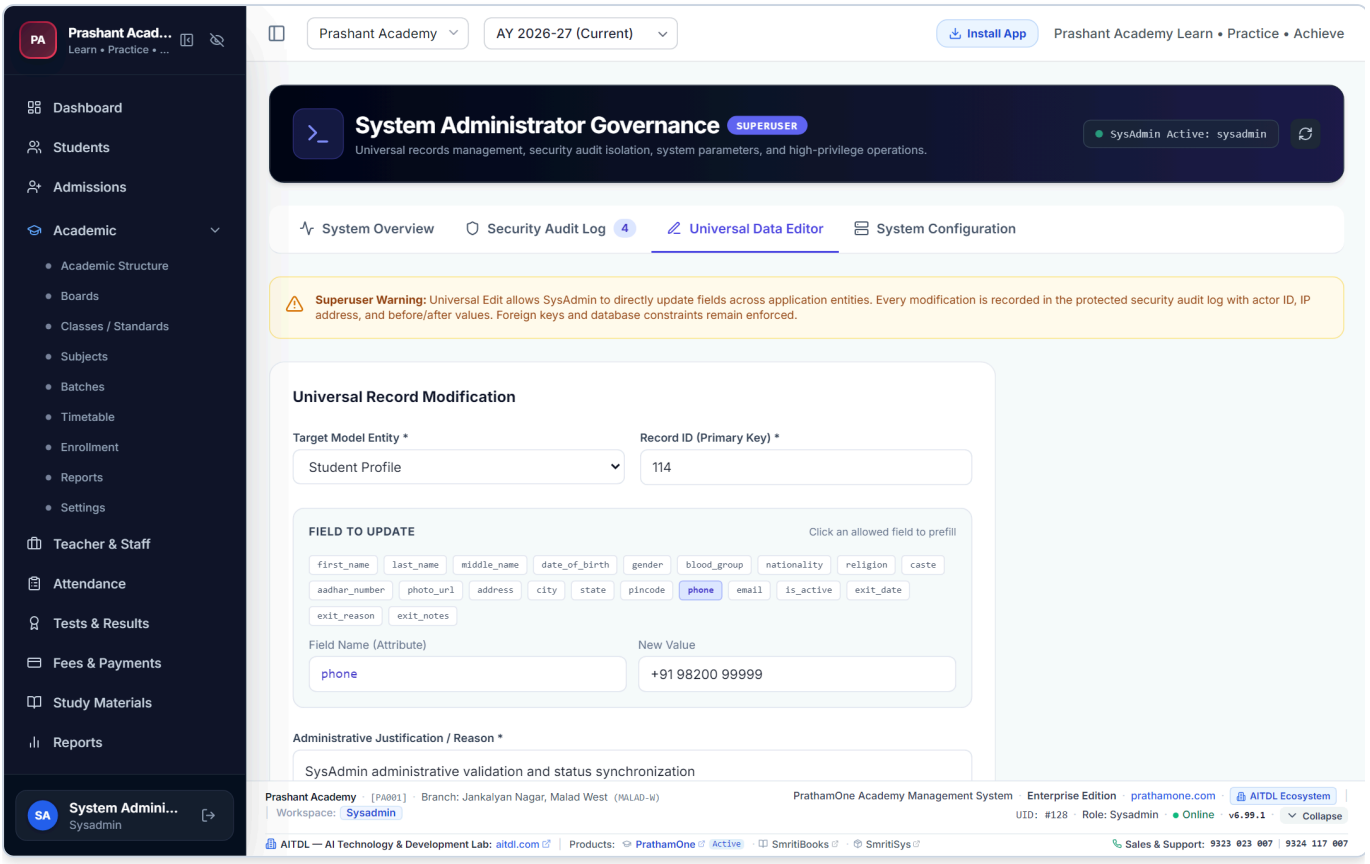
- **Universal Records Editor:** Surgical in-place editing of master data records across 8 core database models with strict allowlist enforcement.
- **Posted Payments Security Gate:** Exclusive authority to unlock and correct posted financial ledger transactions.
- **Immutable Security Audit Logs:** Complete forensic surveillance of all administrative updates, document generation, and authentication events.
- **Multi-Branch Tenant Isolation Management:** Supervision of institutional settings, branch hierarchies, and data isolation boundaries.



SysAdmin Operations Center with Server Diagnostics & Security Logs.

2. Universal Records Editor & Field Allowlists

Located in the **SysAdmin Center** (`/sysadmin`), the Universal Records Editor allows authorized superusers to correct operational anomalies (e.g. spelling errors, wrong exit dates, misallocated contact numbers) directly through the authenticated API without requiring terminal database access.



Universal Records Editor with Permissible Field Chips and Mandatory Audit Reason.

Strict Field-Level Allowlists

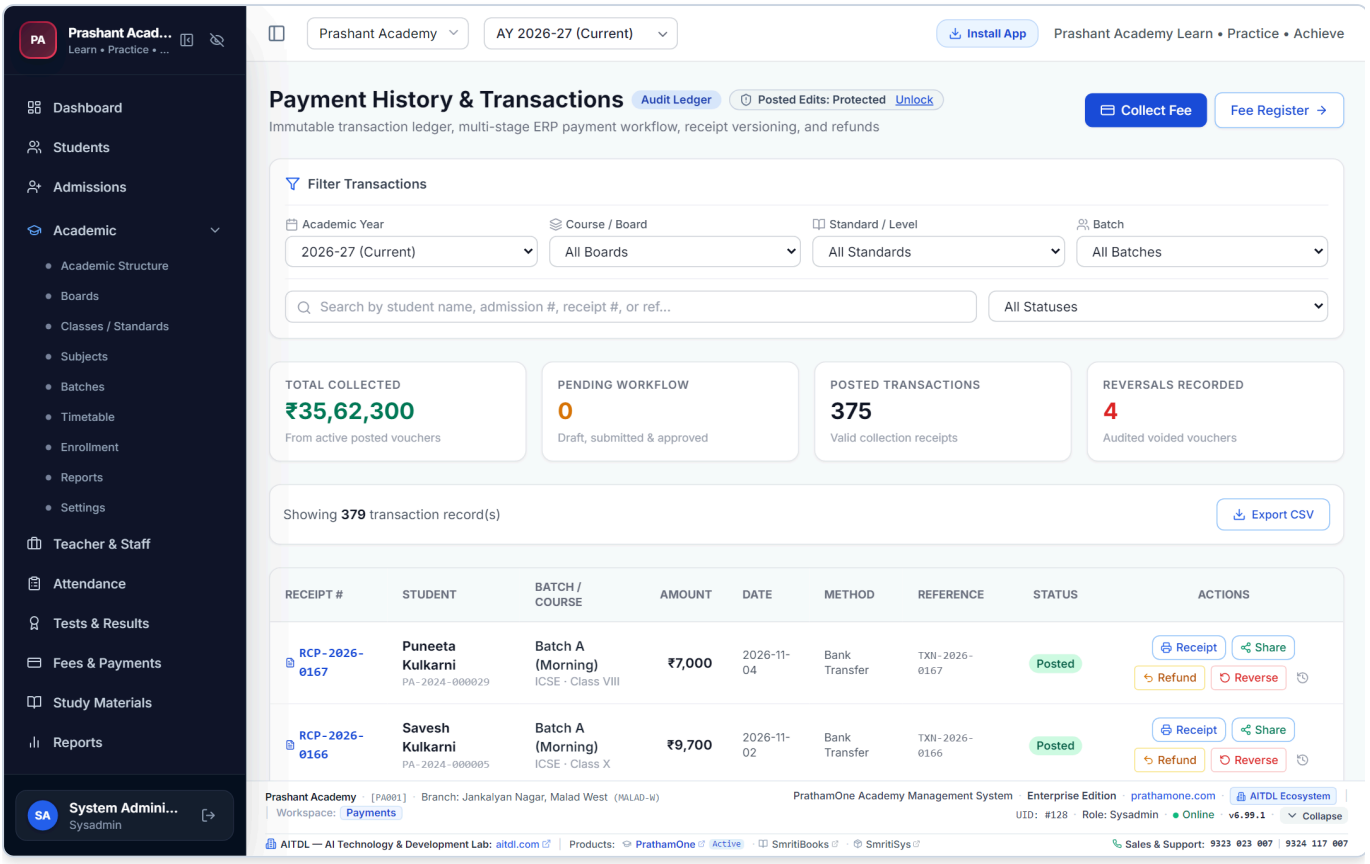
Target Model	Permissible Fields	Prohibited Fields (Blocked by Backend)
`student`	first_name, last_name, middle_name, date_of_birth, gender, blood_group, nationality, religion, caste, aadhar_number, photo_url, address, city, state, pincode, phone, email, is_active, exit_date, exit_reason, exit_notes	id, institution_id, registration_number, created_at, updated_at
`staff`	first_name, last_name, designation, department, qualification, experience_years, phone, email, address, status, joining_date, specialization	id, institution_id, staff_code, created_at, updated_at
`user`	first_name, last_name, email, phone, role, is_active	id, institution_id, username, hashed_password, created_at
`fee_structure`	name, description, total_amount, is_active	id, institution_id, standard_id, created_at
`division / batch`	name, code, room_number, capacity, schedule_timing, status, is_active, class_teacher_id	id, institution_id, standard_id, created_at
`standard`	name, level, description, is_active	id, institution_id, board_id, created_at
`institution`	name, tagline, address, city, state, pincode, phone, email, website, logo_url, favicon_url, primary_color, secondary_color, accent_color, description, is_active	id, code, created_at

Mandatory Reason Requirement

Every universal edit requires an explicit administrative reason. Missing or whitespace reasons are rejected by the backend with **HTTP 400 Bad Request**.

3. Posted Payment Security Gate & Receipt Integrity

Financial ledger security requires that posted transactions cannot be altered by front-desk or accounting personnel. When a legitimate correction is required (e.g. cheque clearance dishonor, wrong fee head mapping), SysAdmin operates the Posted Payment Security Gate:



Posted Payment Security Gate Modal Dialog.

1. **Security Gate Verification**

   SysAdmin enters the payment record ID and authenticates the unlock action.

2. **Mandatory Edit Reason**

   SysAdmin enters a comprehensive justification for the transaction correction.

3. **Receipt Supersession & Audit Recording**

   The backend updates the payment, marks the original receipt as superseded (`is_superseded = True`), and issues a versioned revision. Receipt sequence numbers are never deleted or renumbered.

4. Security Audit Logs & Telemetry

All sensitive operations across PrathamOne EduOS generate structured audit logs stored in the PostgreSQL `audit_logs` table. These can be reviewed in real time via the SysAdmin Center (`/sysadmin` → *Security Audit Logs*):

- **Actor & Identity:** User ID, Username, and Role of the actor.
- **Resource Type & ID:** Target entity (e.g. `student:129`, `payment:45`).
- **Before / After Diff:** Complete JSON snapshot of modified fields (`old_data` vs `new_data`).
- **Origin Network:** Client IP address and browser User-Agent header.

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